A Case Study to Analyze the Impact of Social Media on Video Game Sales

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Abstract **– Social media data is becoming more and more valuable in analyzing patterns in human behavior as a result of the ongoing expansion of online engagement. Twitter is a great place to find current user-generated data. Anyone can utilize this platform to contribute their own ideas, opinions, or feedback on a variety of issues through Tweets. These topics might come from a variety of industries, including business, politics, and entertainment. On average, 6,000 tweets are uploaded on Twitter every second, for a daily total of over 500 million messages. Understanding the causes and impacts of social, environmental, or financial events can be made much easier by the amount of publicly available information. The data is incredibly beneficial for retailers for market research and sales forecasting when paired with cutting-edge data analysis and Artificial Intelligence (AI) solutions. The proposed study examine the sentiments and statistics that have been submitted by users on the Twitter platform in order to determine whether "Social Media Influencing" has any effect on the video game business.**

Keywords – Sentiment Analysis, Social Media, Twitter, Data Analysis

I. INTRODUCTION

Data analytics has the power to drastically change a retail company's operations by providing crucial information to areas like sales, marketing, and logistics. Retailers locate, gather, and evaluate data produced across their departments using a technique called retail analytics. These processes generate a set of useful insights based on performance data, emerging patterns, and business trends. These insights may be used by an organization to enhance key aspects of the retail industry, including customer service, logistics, management, and even sales. With the rise in daily active users on social media, the data and content posted by people on social media platforms such as Twitter can give an organization a lot of data and help them understand the sentiments of the customers.

The buying habits of consumers are continually evolving, creating new opportunities and hazards as consumer tastes change and the retail environment changes. With the data that is now available, it is feasible to understand how consumers engage with e-commerce platforms well. The analysis of user experiences with online shopping is an example of the type of data and visibility accessible which includes seeing which things customers click on and what they purchase at the conclusion of their online shopping. In the actual stores, this visibility is clearly less obvious. Finding a technique to gather information about customer behavior at the retail store and combining it with information about customer behavior online might offer complete 360^{o} view of customer activity.

This study will take a regional approach of analyzing trends by taking into consideration the sales of video games in the US retail market and the Canada retail market and the social media behavior and sentiment of people regarding the games to assess whether there is any correlation between the KPIs taken from the twitter data and video games' sale.

For clients to grow their companies in a changing environment, the NPD Group, L.P. offers data, industry expertise, and prescriptive analytics. The most successful companies in the world rely on NPD to aid in evaluating, forecasting, and maximizing sales performance. For the purposes of this study, the NPD Group has made available point of sale (POS) data on video game sales. The study's baseline data comes from this information and the social media data that was collected, which were based on a 18-week and 15-week time frame respectively for the video games included in this study.

II. RELATED WORK

Businesses in today's world heavily rely on data. Because of the sheer amount of information involved as well as the language used to represent ideas, such as acronyms, memes, and emoticons, Twitter's microblogging data poses serious problems. Sentiment analysis aids in monitoring and deciphering the opinions of the users on Twitter. This gives organizations new opportunities and adds a new dimension to the metrics used to evaluate social media marketing effectiveness. It is a critical component when it comes to customer satisfaction, customer loyalty, branding and promotion, and product acceptability. Understanding the intensity of the emotion represented in words, in addition to the binary polarity (positive or negative), is advantageous for a number of enterprises working in industries that are consumer driven.

Twitter sentiment analysis using a machine learning classifier was suggested by Yadav et al. [1] They used the Kaggle dataset that was made available, which consisted of sentences and keywords relating to a specific product. Positive and negative attitude kinds were meant to be categorized by the authors. For pre-processing, they changed the tweets' case, added spaces where there should have been, eliminated superfluous spaces, and removed quotes and spaces at the conclusion of the tweets. The feature was then extracted using unigrams and bigrams. Lemmatization and stop words were then eliminated. They evaluated the effectiveness of their suggested approach

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using Naive Bayes, Logistic Regression, and Support Vector Machine, three major machine learning classifiers.

During the COVID-19 pandemic in 2020, Machuca et al.[2] presented the sentiment analysis of English tweets. They used English tweets with the hashtag #coronavirus as their data set. By changing all of the letters to lowercase and eliminating all stop words, non-English terms, and punctuation, the data was cleaned. The text data was then transformed into a vector for sentiment classification using a technique called Term Frequency Inverse Document Frequency (TF-IDF). The accuracy of their model's classification of the emotions expressed in the tweets into positive and negative groups using the Logistic Regression technique was 78.5 percent.

Sentiment analysis was done on tweets on social distance in Canada by Shofiya et al. [3] They collected tweets that used the hashtags "#social distancing," "#physical distancing," and variations of those terms. They cleansed the data by eliminating any words with fewer than three characters, punctuation, and URLs. The data was afterwards tokenized, shrunk, and stemmed. After text pre-processing, the feature was extracted from the data using Bag-of-Words, TF-IDF, and Word2Vec. SVM was the classification technique utilized in their paper, and it had an accuracy rate of 87 percent when categorizing tweets as positive or negative.

Recent research suggests that tweets play an important role in prediction of of important events, such as elections and national uprisings. The underlying assumption is that the context, timing, and content of tweets can predict future events. In a paper from the University of Virginia in 2014, the issue of "Can tweets generated by residents of a large U.S. city be employed to anticipate local crime activity?" was raised.. [4] The study shows that the incorporation of features from Twitter enhances prediction accuracy for 19 of 25 crime categories, and it does so significantly for several surveillance horizons.

A rising number of studies have examined Twitter activity during the ongoing COVID-19 worldwide pandemic since January 2020 [5-9]. One researcher looked at the prevalence of 22 different keywords in 50 million tweets between January 22, 2020, and March 16, 2020, including "Coronavirus," "Wuhan," and "Covid-19." Another study used data from Twitter to predict the flow of true and false information to evaluate the themes of English-language tweets from March 10–29, 2020, as well as linguistic distribution and the spread of false news to ascertain how public policy is regarded.

In a study [10], C.J. Hutto et al. used Twitter data directly and applied VADER on it. In addition, they conducted a comparison of VADER, the Valence Aware Dictionary for Sentiment Reasoning, with LIWC, ANEW, SentiWordNet, and the Support Vector Machine (SVM) algorithm. They had tested the attribute chosen using both quantitative and qualitative methods before combining it with five general grammatical rules to determine the sentiment intensity. VADER outperforms the rest of the employed algorithms. However, this VADER is not utilizing any parts of the tweets or words in his writing.

To analyze student evaluations from many teachers of a single source in this paper [11], H. Newman et al. used VADER. Official evaluations, forum comments, and other sources of data have all been used for these evaluation objectives. They compared frequently used comments and will determine the comment's polarity because some questions can be common to all university professors across all faculties. A quick look feature on both positive and negative comments is offered by VADER. As a measurement, the correlation between the comment's overall sentiment score and the class's overall sentiment was used.

By utilizing a sentiment analysis methodology and the data provided by Twitter, P. Singh et al. [12] have examined the impact of the Indian government policy—the demonetization—from the perspective of the average person. Tweets containing the hashtag #demonetization were collected. analysis using geographic location (State wise tweets are collected). The six categories of the sentiment analysis API are joyful, sad, very sad, cheerful, neutral, and no data.

In another study [13], Al Amin et al. discuss their study of Bengali NLP. They changed VADER to support Bengali sentiment identification of polarity to achieve this. They modified the English VADER's capabilities so they could classify Bengali without using a translation. From the experiment, the sentiment analysis of Bengali text was enhanced over the previous model. Additionally, they used the English lexicon's polarity to innovate the Bengali lexicon. To provide a better outcome, they used stemming, a list of Bengali words that are good for you, as well as bigram and trigram applications. They didn't utilize it in sentences that contained both good and negative connotations.

Several studies examined polls using the Twitter API and made statistical predictions about the political outcomes [14], [15]. Here both descriptive and inferential statistics employed. In [14] O'Connor et al. focused on political views and consumer confidence. To understand people's sentiment, correlation analysis, a linear least-squares model, and a regression model were used. To create a polling indicator, the relationship between sentiment ratio and consumer confidence was determined. Additionally, a regression model was used to the forecasting analysis, and the graph produced showed which variables were worse and better predictors of consumer confidence for each part. It suggested that sentiment analysis may be used in numerous situations to measure qualitative phenomena.

Numerous studies have been conducted on the application of AI to the retail sector as well as to AI and social media independently. However, there hasn't been much research done on fusing these two fields with AI techniques, and this paper examines the possible advantages of doing so.

III. METHODOLOGY

A RESTful web service was developed and linked to the Twitter API in order to retrieve and store Tweets. A RESTful web service exposes a set of resources that specify the points of contact with its clients. A GET and

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POST endpoint were created to manage the storing and retrieval of Tweets. Due to the time frame of the sales data, the twitter data needed was collected using the Twitter API for Academic Research which provides access to the entire Twitter archive.

A. Sentiment Analysis

The text sentiment analysis model VADER (Valence Aware Dictionary for Sentiment Reasoning) considers both the polarity (positive/negative) and the degree (strong) of sentiment. For the development of VADER, a combination of qualitative and quantitative approaches were used and then experimentally validate a gold-standard sentiment lexicon that is particularly sensitive to microblog-like conditions. The focus of VADER is social media. VADER therefore spends a lot of time figuring out the sentiments of content that frequently appears on social media, like emoticons, repeating sentences, and punctuation (exclamation marks, for example)

B. Data Pre-Processing

The data pre-processing stage is essential since it affects how well the subsequent stages work. It comprises changing the syntax of tweets, deleting unnecessary information from tweets, and objectifying any other useful features. The following python regular expressions were used for data preprocessing:

- (http|https|ftp)://[a-zA-Z0-9/]+ for removing URLs
- @(w)+ for removing handles
- #(w)+ for removing hashtags

C. Data Sources

Table I shows the data that can be accessed through the tool provided by The NPD Group and Table II shows the features that were collected from the Twitter API.

TABLE I. NPD DATA

Feature	Description	Sample
Ppweek	Week of the year	18
itemDescription	Description of the item	PS4 FIFA 19
observedUnits	Number of units sold	7714
observedValue	Value of units sold	546,789.90

TABLE II. TWITTER DATA

Feature	Description	Sample
lang	Language of tweet	En
source	Platform of tweet	Twitter for Android
created_at	Time and date of tweet	2021-01-19TO5:20:30
text	Content of the tweet	What a beautiful day
country	Location of the tweet	Canada
referenced_tweet	Original tweet information	{text:“”, country: “” ,created_at: “”}

IV. RESULTS

The study's goal was to make a hypothesis about whether and which external KPIs would benefit current AI and analysis tools the most from social media data. Explained below are the results of 2 use cases of popular video games, “Game 1” and “Game 2” over a 18-week and 15-week period. Figure 1 shows results over a 18 week period representing a spike in the sales data for the game in the US Retail market and the number of original tweets about the same game, also originating from the USA. A spike in the number of tweets is observed in week 10 and is followed by a spike in the number of sales for week 12. No spike in tweets was observed before the peak in sales for week 16 and 17. Due to the small change in the number of Tweets produced at the same time, the data's lack of correlation suggested that the sales peak may not have been driven by consumers. However, the correlation in the data for the spike in tweets and sales in weeks 10 and 12 respectively warranted a deeper analysis into the data.

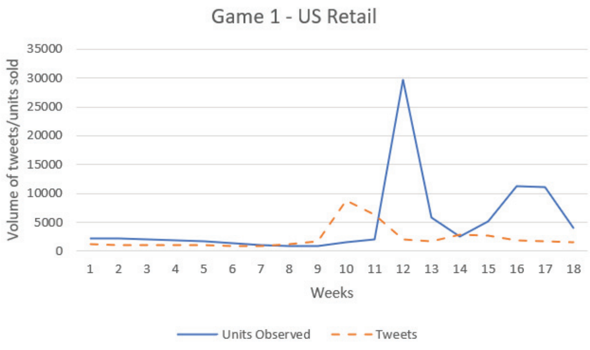

Fig. 1. Units Sold & Number of Tweets - 18 Week Period

Figure 2 shows the results for the same 18 week period, but the number of retweets (original tweets reposted by other users) is also taken into account. Clear spikes in the number of tweets can be seen in week 10 and 15 followed by spikes in sales data in week 12 and 16. The correlation is strenthened by looking at the sentiments of the users when there is a spike in the volume of tweets.

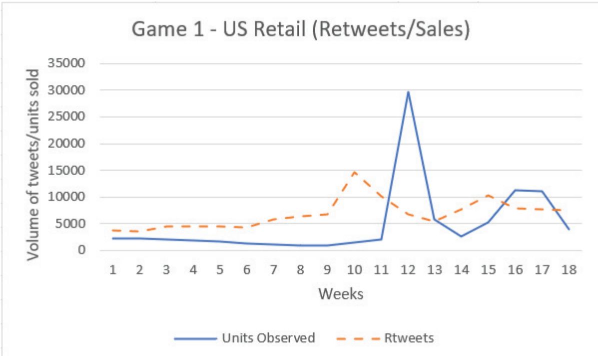

Fig. 2. Units Sold & Number of Retweets - 18 Week Period

When sentiment analysis was conducted on all tweets, the results, displayed in Figure 3, showed mostly neutral sentiments. It is observed in week 10 where there was a spike in volume of tweets, the results were extremely positive at 52%, followed by neutral at 31% and negative at 17%. An increase in the positive sentiment was also observed for week 15 with 45% positive followed by 32% neutral and 23% negative sentiments. On deeper analysis, it was observed that the sudden increase in positive sentiments in week 10 and 15 was actually a gradual change through the previous weeks. It was also observed that after the spike in tweets in week 15, the positive sentiments remained relatively high. Another interesting

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thing to note is that the negative sentiments did not change much and the increase in positive sentiments was coming from the decrease in neutral sentiments.

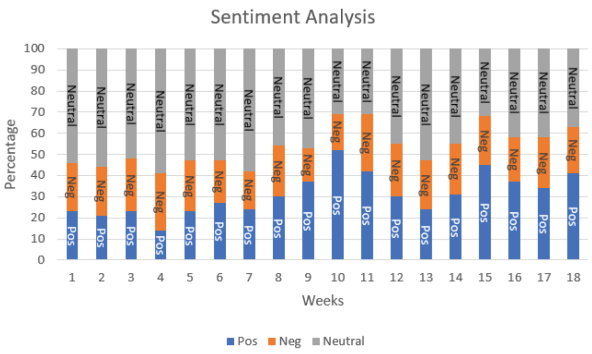

Fig. 3. Sentiment Analysis

Figure 4 shows the results for the same 18 week period, but the sales and twitter data is from Canada. Though the sales for this particular game are relatively low in the Canada retail market, we can see clear spikes in the twitter data, nore specifically in the retweets. In this case, it is observed that the number of original tweets is reducing, however, the number of retweets is increasing.

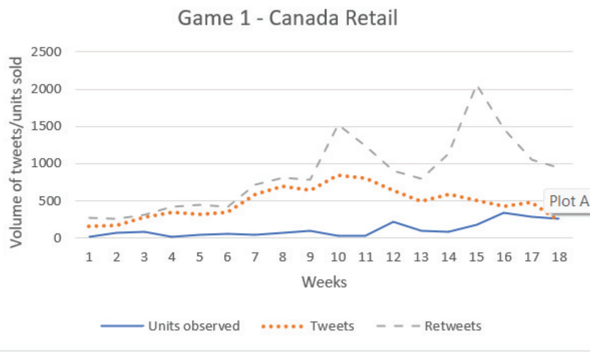

Fig. 4. Units Sold, Tweets & Retweets – 18 Week Period

Figure 5 shows the results for Game 2 in the Canada retail market along with the tweets and retweets from Canada regarding the same game for a 15week period. A pattern is observed where there are distinct spikes in the retweet data followed by spikes in the sales data.

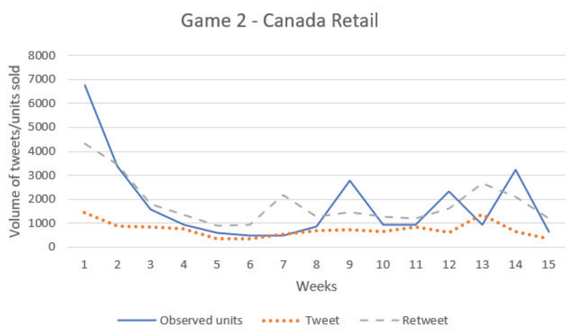

Fig. 5. Units Sold, Tweets & Retweets – 15 Week Period

It is found that "retweets" is an important KPI and deeper analysis into the twitter data from Canada for Game 2 revealed some interesting outcomes which are detailed in Table III. While looking at the twitter data from Canada, during the weeks where a spike in the volume of tweets were recorded, more than 50% of the tweets, while being from Canada, had the source tweet that was originated outside Canada indicating that social media engagement is not bound by geographical barriers.

TABLE III. RETWEET ANALYSIS

Week	% of tweets outside Canada	% of influencer tweets
1	74%	57%
7	51%	34%
11	59%	39%
13	68%	46%

Identified after analyzing the results were the following points:

- Retweets is a very important KPI which should be considered along with original tweets.
- A pattern can be observed where there is a spike in the retweets followed by a spike in the sales data, however analysis over a longer duration of time is required.
- The impact caused by social media, twitter in this case, may not be bound by geographical boundaries i.e., Influencers may cause an impact in other regions as well and not just where they are based in.

Table IV below details the summary of the tweets collected.

TABLE IV. SUMMARY OF TWEETS

Tweets	Volume
Game 1 – Original Tweets (USA Retail)	39304
Game 1 – Tweets + Retweets (USA Retail)	121663
Game 1 – Original Tweets (Canada Retail)	8563
Game 1 – Tweets + Retweets (Canada Retail)	15559
Game 2 – Original Tweets (Canada Retail)	11072
Game 2 – Tweets + Retweets (Canada Retail)	27722

V. CONCLUSION AND FUTURE WORK

One KPI, the volume of Tweets about a certain product, was extracted from social media data and correlated against the sales of that product in a case study spanning an 18-week period and two separate games. This correlation clearly demonstrates the possibility for further investigation. Retweets and normal tweets are included in the KPI. Utilizing AI tools, social media data, other external KPIs, retailer data, and information from these sources can all result in positive effects. This study also revealed that social media's "influence" is not exclusive to one particular region. This is demonstrated by the extremely high number of retweets coming from Canada when the source or initial tweet came from the USA.

Future research will examine the application of novel analysis techniques to understand consumer purchasing trends using contextualized user-driven data and external

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KPIs with the aim of utilizing extracted data provided by NPD. It will also be examined which clustering techniques are most appropriate, such as KMeans, SOM, and Mean-Shift, with the possibility of training models using big historical data sets and testing on the most recent social media and retailer data.

The results observed in this study raise the possibility of developing a metric to determine the "influence" a user has on social media. Retweets has been observed to be quite an important KPI rather than just the original tweets and this, along with other KPIs such a geographical location, followers, likes, engagement, etc. raises the question of whether these KPIs can be utilized to define a metric of social media influence.

The results also raise the possibility of building a dynamic knowledge graph of social media influencers, especially users with high follower counts and high engagement such as comments, likes, reshares, etc. that would clearly show the changing dynamics between influencers and regular users and the impact that the influencers have on the online behavior of the users and the creation and propagation of online trends without the restrictions of geographical boundaries or regions.

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